

JATIN KUMAR

Aspiring AI Engineer & Data Scientist

Greater Noida, Uttar Pradesh, IN | +91 8700496400 | jatin.johnny007@gmail.com | linkedin.com/in/jatinkumar2005 | github.com/jatinkumar300403

Summary

CS major at VIT Vellore, India with hands-on Python experience across machine learning, data cleaning, and pipeline development. Published researcher at ICCS'25 (Springer, Singapore), applying statistical and time-series techniques (Mahalanobis distance, covariance modeling) across multiple structured datasets. Comfortable writing clean, documented, testable Python and building simple APIs/UIs to package analytical tools. Strong academic grounding in ML fundamentals time-series analysis, clustering, NLP gained via coursework, a research internship at IIT-BHU, and independent projects involving large-scale data parsing, cleaning, and retrieval. Eager to apply this foundation to structured signal and log data such as automotive telemetry and diagnostics.

Work Experience

ML Research Intern

Sep 2024 – Apr 2025

IIT-BHU, Varanasi, IN

- Designed and ran controlled experiments across 3 structured molecular datasets, writing Python scripts to preprocess, clean, and validate raw data for accuracy before training.
- Applied statistical/time-series techniques (Mahalanobis distance, covariance modeling, Elastic Weight Consolidation) to reduce forgetting, achieving 90.89%–92.62% accuracy and 0.0–0.27 forgetting, outperforming 10+ SOTA methods.
- Documented methodology for peer review; published as first author at ICCS'25 (Springer LNCS) — [Paper](#)

Education

M.Sc in Computer Science

Oct 2026 – Present

Leibniz Universität Hannover, Hannover, Germany

B.Tech. in Computer Science and Engineering

Sep 2022 – Jul 2026

Vellore Institute of Technology (VIT), Vellore, India

Grade: 1.8

Relevant Courses: Machine Learning, Computer Vision, Artificial Intelligence, Statistics, Algorithms

Projects

1. Kitaab – Semantic Retrieval System

Python, Pandas, LangChain, ChromaDB, OpenAI Embeddings, Hugging Face, Gradio

- Cleaned, filtered, and validated 5,200+ raw text records for accuracy and completeness, then built a retrieval pipeline embedding them into a vector database for top-K similarity search.
- Applied NLP techniques — zero-shot classification and DistilRoBERTa-based sentiment tagging — to enrich and cluster records by category and tone. Packaged behind a Gradio UI. Live at: [Kitaab](#)

2. Aarogyam – AI-based Health and Disease Analyzer

Python, PyTorch, Gemini API, Streamlit

- Built a data pipeline ingesting real-time numerical sensor streams (heart rate, SpO2) from an IoT device, writing scripts to parse, filter, and validate incoming telemetry for downstream analysis.
- Integrated an LLM (Gemini API) via prompt engineering for symptom-based diagnostic reasoning over structured sensor data and unstructured text/image inputs, achieving under 2s inference latency. Delivered via a documented Streamlit UI. Live at: [Aarogyam](#)

3. SwapIT – Add/Drop Portal

TypeScript, Next.js, TailwindCSS

- Built a full-stack scheduling tool for faculty slot swapping at VIT, handling structured request data and validation logic end-to-end. Improved scheduling efficiency by 85%, streamlining 250+ exchanges/semester. Live at: [SwapIT](#)

Technical Skills

Languages: Python, C++, SQL

Machine Learning & Data: NLP, PyTorch, Pandas, NumPy, Scikit-learn, Data Cleaning & Preprocessing

AI/LLM: LangChain, RAG, Vector Databases (ChromaDB), Hugging Face Transformers, Generative AI

Software Engineering: FastAPI, Git/GitHub, Jupyter, React.js/Next.js, MySQL,

Platforms: AWS, Vercel

Languages

English: C1 (Advanced) **German:** A2 (Elementary)

Achievements

- Published Researcher, ICCS'25 (Springer LNCS), Singapore – conference paper on continual learning
- Volunteered as a fund-raiser for Muskurahat Foundation (A child-welfare organization)
- Top 10 Finalist, Yuvamanthan Hackathon 2024, organized by IIT Madras